

AUTONOMOUS QUADRUPED NAVIGATION ROBOT

Technical Project Brief

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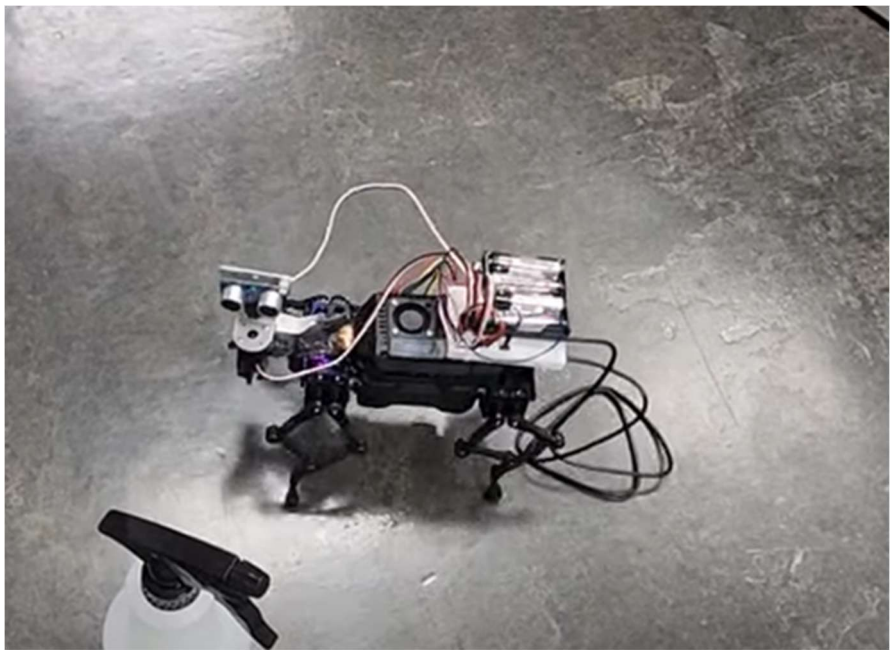
Project objective: design a mobile quadruped robot that travels from one corner of an arena toward the opposite corner while detecting obstacles with external sensors and using programmed logic to change its path.

Project at a Glance

Platform WAVEGO 12-DOF quadruped	Host computer Raspberry Pi 5	Lower-level controller ESP32	Final sensor Servo-mounted PING))) ultrasonic sensor
Original sensors TFMini-S LiDAR + QMC5883P magnetometer	Navigation Obstacle avoidance + wall detection/following	Libraries JSON, RPi.GPIO, serial + time	Robot motion Existing ESP32 gait control

1. System Architecture

The WAVEGO kit used an ESP32 for leg movement, gait control and inverse kinematics. A Raspberry Pi 5 served as the host computer. Because the ESP32 already contained gait-control software, the project focused on external sensor integration and navigation logic running on the Raspberry Pi rather than modifying the lower-level gait code.



WAVEGO 12-DOF platform used during the project.

2. Sensor Revision

The original design called for a TFMini-S Micro LiDAR sensor and an Adafruit QMC5883P three-axis magnetometer. The report states that both devices proved difficult to implement through Python on the Raspberry Pi. The LiDAR was replaced with a PING))) ultrasonic distance sensor mounted on a 180-degree servo so the sensor could sweep through multiple angles and scan for obstacles.

Removing the compass also required a change in navigation strategy. The final logic used two pathing phases: obstacle avoidance while moving toward a wall, then wall following toward the target corner.

3. Navigation and Software Logic

- distance() measured ultrasonic distance and returned the value in centimeters.
- set_angle(angle) converted the requested servo angle to PWM duty cycle for sensor positioning.
- Avoid() handled obstacle-avoidance logic and used elapsed avoidance time to determine whether the detected object should be treated as a wall.
- turn(direction) sent direction values through the robot gait-control interface, with -1 used for clockwise rotation and 1 used for counterclockwise rotation.

The main const_sweep routine moved the sensor through its sweep in 4-degree increments while taking distance readings. When an object was detected, the routine called the avoidance logic. After wall detection, the software transitioned to the second navigation phase and continued until the next wall was detected in front of the robot, at which point a stop command was sent.

4. Testing and Troubleshooting

- Battery life limited the number of consecutive trials available during troubleshooting.
- The team observed ESP32 errors and inconsistent leg-servo behavior during some testing periods.
- Raspberry Pi-to-ESP32 communication required trial and error because multiple vendor code versions were available.
- Wi-Fi issues sometimes led to reflashing the Raspberry Pi and rewriting code that had been lost.

5. Engineering Changes Documented in the Report

The clearest design change documented in the report was the move away from LiDAR and magnetometer sensing to a servo-scanned ultrasonic sensor. That hardware change also required the team to replace compass-based navigation with obstacle avoidance, timed wall detection and wall-following logic.